

PROJECT SYNOPSIS

DAD — Deepfake Audio Detection System

A Hybrid CNN + LSTM + Transformer Framework for Real-World Synthetic Speech Detection

Final Year Project — Bachelor of Technology
Department of Computer Science & Engineering

1. Title

Full Title: DAD — Deepfake Audio Detection System: A Hybrid Deep Learning Framework Using Convolutional Neural Networks, Long Short-Term Memory Networks, and Transformer Encoders for Robust, Real-World Synthetic Speech Detection with Error-Driven Fine-Tuning

2. Introduction

2.1 Background and Domain Overview

The rapid and largely unregulated advancement of generative artificial intelligence has precipitated a paradigm shift in speech synthesis technology. Over the last decade, the progression from concatenative speech synthesis and statistical parametric models to modern end-to-end neural text-to-speech (TTS) architectures has been staggering. Contemporary neural TTS systems — including Tacotron 2, FastSpeech 2, VITS, XTTS, and VALL-E — are capable of producing synthetic speech that is, in controlled listening tests, perceptually indistinguishable from recordings of authentic human speech. Similarly, neural voice conversion (VC) systems can transplant the vocal identity of a target speaker onto an arbitrary source utterance, effectively enabling the synthesis of any person's voice from as little as three to five seconds of reference audio.

These capabilities, while legitimate and valuable for applications in accessibility technology, entertainment, dialogue systems, and language learning, simultaneously represent a profound and growing threat to the integrity of audio communication. The term deepfake audio refers broadly to any synthetic or manipulated speech signal produced using AI-based methods with the intent to deceive listeners about the true origin or content of the recording. Unlike visual deepfakes — which are subject to increasing public scrutiny and media literacy — deepfake audio operates with greater subtlety, is easier to produce at scale, and is significantly harder to detect without automated systems.

2.2 Problem Statement and Motivation

The societal consequences of undetected deepfake audio are already materialising. In 2019, attackers used AI-cloned voice audio to impersonate the CEO of a UK-based energy company, successfully authorising a fraudulent wire transfer of EUR 220,000 in a single phone call. Political disinformation campaigns have exploited synthetic audio to fabricate statements attributed to public figures. Courts and forensic investigators in multiple jurisdictions are now confronting the question of audio evidence authenticity. Voice-based authentication systems — deployed by banks, call centres, and government services — face a direct vulnerability to voice cloning attacks. These incidents collectively underscore the urgent and pressing need for reliable, automated deepfake audio detection systems that can operate across diverse acoustic conditions, speaker identities, and synthesis technologies.

The DAD (Deepfake Audio Detection) system is proposed in direct response to this need. It seeks to deliver a production-grade detection pipeline that achieves robust real-world generalisation, not merely high performance on constrained academic benchmarks. The project's central thesis is that genuine progress in this domain requires not only architectural innovation but also rigorous data engineering, diversity-aware multi-dataset training, and error-driven model refinement — a thesis supported by the empirical results presented in later sections.

2.3 Industry and Academic Relevance

Academic interest in deepfake audio detection was formalised through the ASVspoof challenge series, initiated in 2015 and continued through 2019 and 2021 editions. These challenges have

produced valuable benchmark datasets and driven significant algorithmic progress. However, the gap between academic benchmark performance and real-world deployment accuracy remains substantial. Industrial organisations including Microsoft, Meta AI, and Google DeepMind have published research on synthetic speech detection, reflecting the commercial urgency of the problem. The European Union's AI Act and the United States' DEEPFAKES Accountability Act both reference the threat posed by synthetic media, including audio, signalling regulatory attention that will increase demand for certified detection technologies.

3. Problem Statement

3.1 Core Technical Problem

The fundamental technical problem addressed by this project is the reliable binary classification of audio samples as either authentic (real) human speech or synthetically generated (fake) speech, under conditions that reflect the diversity and complexity of real-world audio environments. This problem is characterised by three interacting challenges:

- **Generalisation Gap:** Existing detection systems trained on single academic datasets — primarily the ASVspoof 2019 LA corpus — exhibit strong benchmark performance but fail to generalise to novel synthesis systems, unseen speaker identities, or differing acoustic conditions. The gap between test-set accuracy on benchmark data and actual deployment performance is often dramatic.
- **High-Quality Voice Clone Detection:** Modern neural TTS systems produce synthetic speech with spectral and prosodic characteristics closely approximating natural human voice. Traditional detection features (e.g., MFCCs, CQCCs) and early deep learning architectures trained on older vocoder-based fakes fail to detect these high-quality clones, as they lack the obvious artifacts on which those models rely.
- **False Positive Problem:** Many detection systems exhibit a bias toward classifying audio as fake, resulting in high false positive rates on authentic speech — particularly clean, studio-quality recordings. This limits practical deployment in contexts where wrongly flagging legitimate audio has significant consequences, such as legal proceedings or identity verification systems.

3.2 Why This Problem is Critical

Socially, undetected deepfake audio enables fraud, harassment, political manipulation, and the erosion of trust in audio communication as a whole. Technically, the absence of generalised detection systems represents an unresolved open problem at the intersection of signal processing, machine learning, and digital forensics. Industrially, organisations deploying voice-based authentication or relying on audio evidence face direct operational and legal risk. Addressing this problem is therefore not merely an academic exercise but a matter of practical urgency with measurable societal impact.

4. Objectives of the Project

The DAD project pursues the following six specific, measurable objectives:

- **Objective 1 — Multi-Source Data Engineering:** Design and implement a 7-phase audio data engineering pipeline that ingests audio from five diverse datasets (ASVspoof 2019 LA, WaveFake, LibriSpeech, Common Voice, VoxCeleb), applies controlled contribution weighting, augments real audio, and produces a balanced, speaker-safe training corpus with approximately equal real and fake sample counts.

- **Objective 2 — Robust Feature Extraction:** Extract Log-Mel Spectrogram features (shape: 64×501 , dtype: float32) from all audio clips and apply Welford streaming z-score normalisation to produce numerically stable, dataset-wide normalised representations without requiring full-dataset memory loading.
- **Objective 3 — Hybrid Deep Learning Architecture:** Design, implement, and train a hybrid neural model combining a CNN (spatial frequency feature extraction), Bidirectional LSTM (temporal sequence modelling), and Transformer Encoder (global multi-head self-attention), followed by a fully connected classification head with sigmoid activation output.
- **Objective 4 — Quantified Performance Targets:** Achieve a minimum ROC-AUC of 0.95 and an F1-score in the range of 0.88–0.90 on the held-out test set, with Recall $\geq$ 0.90 and Precision improvement above the baseline value of 0.7830.
- **Objective 5 — Error-Driven Fine-Tuning:** Implement a 10-phase fine-tuning pipeline incorporating hard negative mining, SpecAugment, decision threshold optimisation over the range [0.3, 0.8], loss rebalancing with pos_weight tuning and label smoothing, and exposure to high-quality modern neural TTS outputs.
- **Objective 6 — Modular and Extensible System:** Produce a modular, well-documented system suitable for integration into downstream applications including media verification platforms, voice authentication anti-spoofing layers, and digital forensic audio analysis tools.

5. Scope of the Project

5.1 System Coverage

The DAD system encompasses the complete pipeline from raw audio input to binary classification output. It processes monaural audio files of arbitrary duration, standardised to 16 kHz and segmented into 5-second clips. The system targets both traditional spoofing attacks — waveform concatenation, signal processing vocoders — and modern neural synthetic speech generated by neural TTS and voice conversion systems. The detection granularity is clip-level binary classification (real/fake) with an associated confidence score derived from the sigmoid output probability.

The system has been designed and validated on five publicly available datasets spanning diverse speaker demographics, acoustic conditions, recording environments, and synthesis technologies. The training strategy enforces speaker-safe partitioning, ensuring that speaker identity does not leak across train, validation, and test splits.

5.2 Target Users

The DAD system is designed to serve the following primary user categories:

- Media organisations and fact-checking bodies requiring automated audio authenticity verification for broadcast content, social media audio, and journalistic evidence.
- Law enforcement and digital forensic practitioners who need to assess the authenticity of audio recordings submitted as evidence in criminal or civil proceedings.
- Cybersecurity professionals and organisations assessing the vulnerability of voice-based authentication and biometric identification systems to voice cloning attacks.
- Academic researchers in anti-spoofing, speaker verification, computational audition, and digital forensics who require a well-documented, multi-dataset baseline system for comparative evaluation.

- Technology developers building downstream applications that require a reliable audio authenticity verification component.

5.3 Current Limitations

The current version of the DAD system has the following acknowledged limitations: (i) Real-time streaming inference is not supported in this release, though the model architecture is compatible with sliding-window implementation; (ii) The system has been trained and evaluated predominantly on English-language audio and may exhibit reduced performance on non-English speech; (iii) Adversarially crafted audio — samples specifically engineered to evade the classifier — has not been evaluated; (iv) Temporal localisation of synthesised segments within longer recordings is not performed; the system outputs a single binary label per 5-second clip.

6. Literature Review

6.1 GMM-Based Systems and Handcrafted Features

The earliest systematic research in spoofed speech detection predates the availability of neural synthesis, focusing on replay attacks and signal-processing-based vocoders. Todisco et al. [2] proposed Constant-Q Cepstral Coefficients (CQCCs) as a front-end feature for GMM-based anti-spoofing classifiers, demonstrating superior performance over MFCCs and LFCCs on the ASVspoof 2015 and 2019 datasets. CQCC features capture fine-grained spectral characteristics sensitive to the phase discontinuities introduced by vocoder synthesis. The CQCC-GMM baseline system for ASVspoof 2019 LA achieved an Equal Error Rate (EER) of approximately 9.57%, establishing an important reference point. However, GMM-based systems are fundamentally limited by their fixed, handcrafted feature representations: they cannot adapt to the continuously evolving landscape of neural synthesis technologies and require extensive manual feature engineering for each new attack type.

6.2 Light CNN (LCNN) Systems

The introduction of deep learning to anti-spoofing research produced substantial performance improvements. The Light Convolutional Neural Network (LCNN), adapted for anti-spoofing by Wu et al. [3], replaces standard convolutional activations with Max-Feature-Map (MFM) activations, enabling the model to learn competitive inhibition between feature maps and producing sparser, more discriminative representations. LCNN systems trained on Log-Mel Spectrogram or LFCC inputs consistently ranked among top-performing entries in ASVspoof evaluations, with EERs below 5% on the 2019 LA evaluation set for some configurations. Despite this, LCNN systems trained on ASVspoof data exhibit brittleness on out-of-distribution audio: when evaluated on WaveFake — which contains modern neural vocoder outputs — LCNN performance degrades substantially, as reported by Frank et al. [6].

6.3 RawNet2 and End-to-End Waveform Processing

Tak et al. [4] introduced RawNet2, a landmark end-to-end architecture that operates directly on raw waveform samples, learning sinc-based filterbanks from data rather than relying on fixed spectral features. RawNet2 consists of sinc-conv filters, residual blocks with feature map scaling, a GRU layer, and a fully connected classification head. The model achieved an EER of 1.12% on the ASVspoof 2019 LA evaluation partition, representing a major advance over spectrogram-based approaches. The key insight of RawNet2 is that fixed spectral features may inadvertently discard or smooth over the fine-grained temporal and phase-level artifacts introduced by neural vocoders, which raw waveform processing can capture. However, raw waveform models are sensitive to the acoustic characteristics of training data and exhibit

significant performance degradation on audio with background noise or channel effects — a limitation with direct relevance to real-world deployment.

6.4 Graph Neural Networks — AASIST

Jung et al. [5] proposed AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks), which represents the current state of the art on the ASVspoof 2019 LA benchmark. AASIST models the heterogeneous relationship between spectral and temporal audio artifacts using a graph attention network in which nodes represent spectral and temporal feature tokens, and edges represent their learned mutual interactions. The model achieved an EER of 0.83% on the ASVspoof 2019 LA evaluation set, substantially outperforming all prior approaches. Despite this benchmark superiority, AASIST is a highly specialised architecture that is difficult to adapt for multi-dataset training or incremental fine-tuning, and its computational complexity limits deployment in resource-constrained environments.

6.5 WaveFake — The Generalisation Challenge

Frank et al. [6] published the WaveFake dataset alongside a comprehensive evaluation of existing detection systems on synthetic audio generated by seven modern neural vocoders: MelGAN, Multi-Band MelGAN, HiFi-GAN, Parallel WaveGAN, WaveGlow, WaveNet, and PWGAN. Their results revealed that models trained exclusively on ASVspoof 2019 data showed dramatic EER increases when evaluated on WaveFake outputs from vocoders not represented in training, with some models performing near-random. This finding constitutes the strongest empirical evidence for the generalisation gap in anti-spoofing research and directly motivates the multi-dataset training strategy adopted in the DAD system.

6.6 Identified Research Gaps

A systematic review of the literature reveals three consistent and significant gaps that the DAD project addresses: (i) Single-dataset training — the dominant paradigm in published work — produces models that fail to generalise across synthesis technologies; (ii) Error-driven or curriculum learning strategies, which selectively expose the model to its hardest misclassified samples during fine-tuning, have received virtually no attention in the anti-spoofing literature despite their demonstrated utility in related domains; (iii) The false positive problem — authentic speech being wrongly classified as fake — has been systematically neglected, with research optimising primarily for EER or recall without adequately penalising false positives. The DAD system's design explicitly targets all three gaps.

7. Proposed System

7.1 System Overview

The DAD system is a complete, end-to-end deepfake audio detection framework comprising five tightly integrated subsystems: a multi-source data engineering pipeline, a feature extraction and normalisation module, a hybrid deep learning classification model, a 10-phase error-driven fine-tuning pipeline, and an evaluation and threshold optimisation module. The system is designed around the central principle that robust real-world detection performance is primarily a function of data quality, diversity, and pipeline engineering — rather than architectural complexity alone.

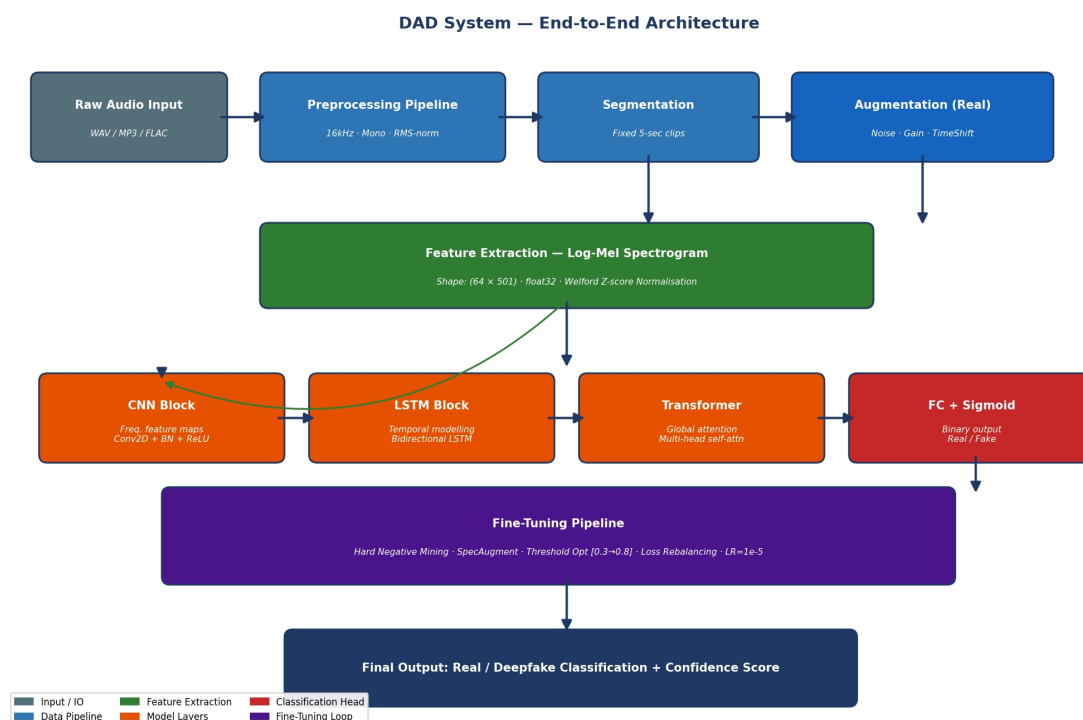


Figure 1: DAD System — Complete End-to-End Architecture

7.2 Multi-Dataset Training Strategy

The dataset composition was designed to provide maximum coverage of both authentic speech variation and synthetic speech diversity. The fake audio corpus draws from ASVspoof 2019 LA (traditional vocoder-based synthesis covering 19 distinct spoofing algorithms) and WaveFake (modern neural vocoder outputs from 7 architectures). The real audio corpus draws from LibriSpeech clean-100 (read speech, high quality), Mozilla Common Voice (crowd-sourced, diverse noise conditions and accents), VoxCeleb1 subset (conversational real-world speech from celebrities), and custom augmented recordings. Controlled contribution weighting prevents any single dataset from dominating the training distribution, ensuring that the learned representations are not biased toward the acoustic characteristics of any one source.

7.3 Key Improvements Over Existing Systems

Relative to the reviewed prior work, the DAD system introduces several substantive improvements: multi-dataset training produces broader coverage of attack types and authentic speech conditions; the hybrid CNN+LSTM+Transformer architecture captures spatial frequency features, temporal dynamics, and global context simultaneously; the error-driven fine-tuning pipeline enables iterative improvement from systematic misclassification — a capability absent from all reviewed systems; and explicit attention to the false positive problem through hard negative mining on authentic audio distinguishes this work from the recall-optimised approaches prevalent in the literature.

8. Methodology

8.1 Development Approach

The project follows an iterative development methodology adapted for machine learning system engineering. Unlike waterfall approaches, the iterative model permits progressive refinement of both data and model components based on empirical evaluation at each stage. Development proceeded through three major iterations: (i) baseline dataset construction and

initial model training; (ii) evaluation-driven identification of failure modes; and (iii) targeted fine-tuning to address those failure modes. This structure reflects the inherently empirical nature of deep learning system development, where the optimal configuration cannot be determined analytically in advance.

8.2 Data Pipeline — 7 Phases

The data engineering pipeline was designed and executed in seven sequential phases, each with specific inputs, processing steps, and validation criteria.

DAD — 7-Phase Data Engineering Pipeline

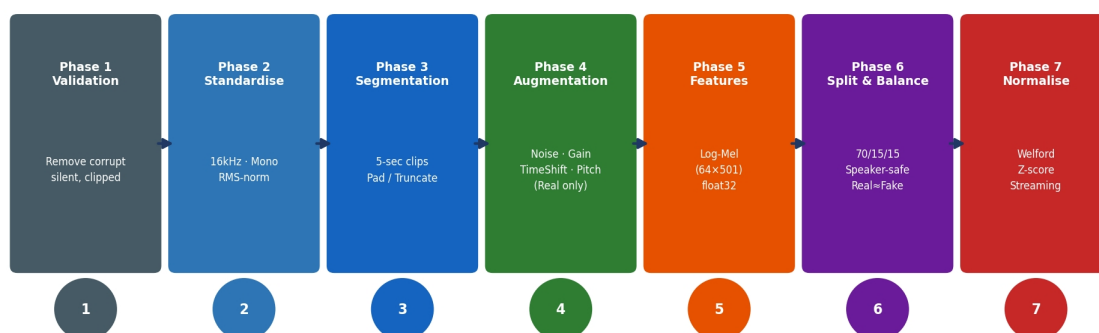


Figure 2: DAD — 7-Phase Data Engineering Pipeline

- **Phase 1 — Validation:** All audio files across all datasets are scanned for corruption (file integrity check), silence (energy below threshold), clipping (sample values saturating at ± 1.0), and format inconsistency. Invalid files are logged and excluded before any further processing.
- **Phase 2 — Standardisation:** All audio is resampled to 16,000 Hz using a high-quality sinc resampler, converted to monaural by averaging stereo channels where applicable, and RMS-normalised to ensure consistent signal level across recordings from different environments and recording chains.
- **Phase 3 — Segmentation:** All audio files are segmented into fixed 5-second clips. Files shorter than 5 seconds are zero-padded at the end; files longer are split into non-overlapping segments with the final segment padded if necessary. This ensures a uniform input dimension for all downstream processing.
- **Phase 4 — Augmentation (Real Audio Only):** To simulate real-world acoustic variability and reduce the model's susceptibility to the clean-audio-equals-fake bias, augmentation is applied exclusively to real audio samples. Augmentation operations include additive Gaussian noise injection (SNR range: 10–40 dB), random gain variation (± 6 dB), time shift (up to ± 0.5 s), mild pitch shifting (± 2 semitones), and time stretching (rate: 0.9–1.1 $\times$). Each real clip is augmented to produce one additional variant, effectively doubling the real corpus size.
- **Phase 5 — Feature Extraction:** Log-Mel Spectrograms are computed for all clips using 64 Mel filter banks, a 25ms Hanning window, 10ms hop length, and 1024-point FFT. The resulting spectrogram shape is (64, 501) — 64 frequency bins $\times$ 501 time frames — stored as float32 arrays. Log compression is applied to suppress the dynamic range of the spectrogram and produce more Gaussian-distributed features.
- **Phase 6 — Dataset Splitting and Balancing:** The dataset is partitioned into train (70%), validation (15%), and test (15%) subsets using speaker-safe stratification: all clips from a given speaker are assigned exclusively to one split. Real and fake sample

counts are balanced within each split to prevent class imbalance from biasing the loss function.

- **Phase 7 — Normalisation:** Z-score normalisation (zero mean, unit variance) is applied globally across all features. Mean and variance statistics are computed using Welford's online algorithm, which processes samples sequentially without requiring the full dataset to reside in memory — a critical consideration for datasets exceeding available RAM.

8.3 Model Architecture — CNN + LSTM + Transformer

The hybrid model architecture is designed to exploit complementary inductive biases of three distinct network families, each contributing a different type of representational abstraction from the input spectrogram.

DAD — Hybrid Model Architecture (CNN + LSTM + Transformer)



Figure 3: DAD — Hybrid Model Architecture (CNN + LSTM + Transformer)

CNN Block: The input Log-Mel Spectrogram (shape: $1 \times 64 \times 501$) is processed by three convolutional layers with 32, 64, and 128 filters respectively, each followed by batch normalisation and ReLU activation. Max-pooling layers after each convolution reduce the spatial dimensions progressively. The CNN's receptive field captures localised frequency-domain patterns: spectral artifacts, harmonic irregularities, and formant discontinuities that distinguish synthetic from natural speech.

LSTM Block: The CNN feature maps are reshaped into a temporal sequence and passed to a 2-layer bidirectional LSTM with 256 hidden units per direction. Bidirectionality enables the model to incorporate both past and future temporal context at each timestep, which is important for capturing the prosodic and rhythmic patterns of speech. Dropout ($p=0.3$) is applied between LSTM layers to prevent overfitting. The LSTM output sequence is mean-pooled to produce a fixed-length utterance-level representation.

Transformer Encoder Block: The pooled LSTM output, along with the full sequence of LSTM hidden states, is passed to a Transformer Encoder consisting of a single layer with 8 attention heads and a feed-forward dimension of 1,024. The multi-head self-attention mechanism attends globally across all timesteps simultaneously, capturing long-range dependencies that sequential LSTM processing may partially obscure. This is particularly valuable for detecting synthesis artifacts that manifest as subtle global consistency violations across an utterance.

Fully Connected Classification Head: The Transformer encoder output is passed through a linear layer followed by ReLU activation and dropout, then a final linear layer with sigmoid

activation that produces a scalar probability score $p \in [0, 1]$, where values above the optimised decision threshold τ are classified as fake and values below as real.

8.4 Training Configuration

Training employs Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss), which combines the sigmoid activation with the cross-entropy loss computation in a numerically stable manner. The AdamW optimiser is used with an initial learning rate of 1×10^{-3} and weight decay of 1×10^{-2} for L2 regularisation. A ReduceLROnPlateau scheduler monitors validation F1-score and reduces the learning rate by a factor of 0.5 when improvement stagnates over 3 consecutive epochs. Automatic Mixed Precision (AMP) training is enabled via torch.cuda.amp, reducing GPU memory consumption and accelerating training by approximately 1.6×. Early stopping with a patience of 10 epochs on validation F1-score prevents overfitting.

8.5 Fine-Tuning Pipeline — 10 Phases

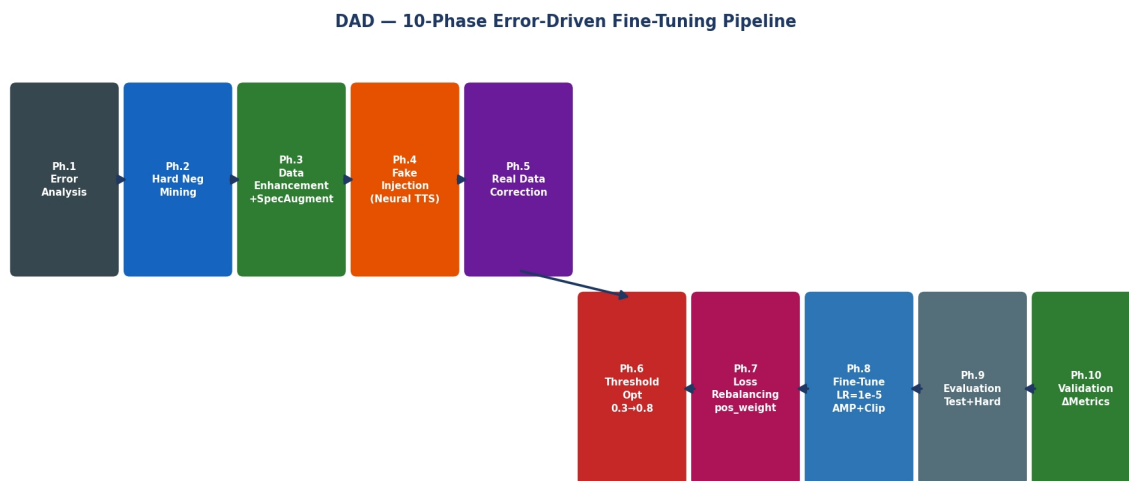


Figure 4: DAD — 10-Phase Error-Driven Fine-Tuning Pipeline

Following initial training, the fine-tuning pipeline iteratively refines the model using an error-driven learning strategy:

- **Phase 1 — Error Analysis:** The trained model is applied to the full training set. All misclassified samples — false positives (real classified as fake) and false negatives (fake classified as real) — are identified and extracted.
- **Phase 2 — Hard Negative Mining:** A fine-tuning subset is constructed by combining all misclassified samples with a balanced selection of correctly classified samples, weighted to emphasise the hardest cases. This subset focuses the fine-tuning gradient updates on the examples where the model is most uncertain or systematically wrong.
- **Phase 3 — Data Enhancement with SpecAugment:** SpecAugment [9] is applied to hard samples during fine-tuning. SpecAugment applies random time masking (masking consecutive time frames) and frequency masking (masking consecutive frequency channels) to the input spectrograms, forcing the model to rely on distributed features rather than specific spectral or temporal regions.
- **Phases 4 & 5 — Fake Injection and Real Correction:** High-quality synthetic audio from modern TTS systems (XTTS, VALL-E variants, ElevenLabs-style outputs) is injected into the fine-tuning set to ensure exposure to the most challenging contemporary synthesis technologies. Simultaneously, additional clean, high-quality real audio is added to reduce false positives on authentic speech.

- **Phase 6 — Threshold Optimisation:** The decision threshold τ is optimised via grid search over the range $[0.3, 0.8]$ in steps of 0.002. The objective function maximises the F1-score on the validation set subject to a minimum recall constraint of 0.88, ensuring that sensitivity to fake audio is not sacrificed for precision gains.
- **Phases 7–8 — Loss Rebalancing and Fine-Tuning Training:** BCEWithLogitsLoss pos_weight is tuned to upweight false negative errors relative to false positives. Label smoothing ($\epsilon=0.1$) is applied to prevent overconfidence. Fine-tuning proceeds at a reduced learning rate of 1×10^{-5} with gradient clipping (max norm: 1.0) and AMP enabled.
- **Phases 9–10 — Evaluation and Validation:** The fine-tuned model is evaluated on both the held-out test set and the hard sample subset. Performance delta ($\Delta F1$, $\Delta Precision$, $\Delta Recall$) relative to the initial model is computed to confirm that fine-tuning produced meaningful improvement without regression on easy samples.

9. Technology Used

- **Programming Language:** Python 3.10+ (primary language for all pipeline, model, and evaluation code)
- **Deep Learning Framework:** PyTorch 2.0+ with CUDA support; torch.cuda.amp for Automatic Mixed Precision; torch.nn for model construction; torch.optim for AdamW and schedulers
- **Audio Processing:** Librosa 0.10+ (feature extraction, resampling, augmentation); SoundFile (audio I/O); torchaudio (SpecAugment, resampling, additional transforms)
- **Feature Extraction:** Log-Mel Spectrograms (64 Mel bins, 25ms window, 10ms hop, 1024-pt FFT) via Librosa; Welford streaming normalisation implemented in NumPy
- **Data Engineering:** NumPy (array operations); Pandas (metadata management, CSV I/O); scikit-learn (train/val/test splitting, metrics: ROC-AUC, F1, precision, recall, confusion matrix)
- **Model Components:** CNN: torch.nn.Conv2d + BatchNorm2d + ReLU + MaxPool2d; LSTM: torch.nn.LSTM (bidirectional); Transformer: torch.nn.TransformerEncoderLayer + TransformerEncoder; FC Head: Linear + ReLU + Dropout + Linear + Sigmoid
- **Training Utilities:** BCEWithLogitsLoss; AdamW; ReduceLROnPlateau; torch.cuda.amp.GradScaler; gradient clipping via torch.nn.utils.clip_grad_norm_
- **Datasets:** ASVspoof 2019 LA; WaveFake (7 neural vocoders); LibriSpeech clean-100; Mozilla Common Voice (multi-accent); VoxCeleb1 subset (conversational)
- **Experiment Management:** JSON-based metrics logging (train/val/test loss, F1, ROC-AUC per epoch); checkpoint saving via torch.save; optional integration with Weights & Biases (wandb)
- **Version Control:** Git with GitHub remote repository; structured commit history by pipeline phase
- **Backend API (Optional):** FastAPI + Uvicorn for REST inference endpoint; pydantic for request validation; returns JSON with prediction label and confidence score
- **Visualisation:** Matplotlib + Seaborn (confusion matrices, ROC curves, training curves, spectrogram visualisation)

10. System Requirements

10.1 Hardware Requirements

- GPU: NVIDIA GPU with minimum 8 GB VRAM (e.g., RTX 3070/3080) for single-GPU training; RTX 4090, A100, or V100 recommended for faster training with full dataset. Cloud GPU instances (Google Colab Pro+, AWS p3.2xlarge, Lambda Labs) are fully supported.
- CPU: Modern multi-core processor with minimum 8 physical cores recommended for parallel data preprocessing using Python multiprocessing.
- RAM: Minimum 16 GB system RAM; 32 GB strongly recommended to support concurrent data loading, augmentation, and model training without swapping.
- Storage: Minimum 200 GB SSD (NVMe preferred for I/O throughput); accommodates raw dataset audio files (~80 GB), processed feature arrays (~60 GB), model checkpoints, and experiment artefacts.
- Network: High-bandwidth internet connection for initial dataset download (LibriSpeech ~6 GB, Common Voice ~55 GB English subset, ASVspoof 2019 LA ~25 GB, WaveFake ~70 GB).

10.2 Software Requirements

- Operating System: Ubuntu 20.04 LTS or 22.04 LTS (primary development and production environment); Windows 10/11 with WSL2 is functionally supported; macOS is supported for CPU-only development.
- Python: Version 3.10 or higher (Python 3.11 recommended for performance improvements).
- CUDA Toolkit: Version 11.7 or higher; CUDNN 8.5+ for optimised convolution operations.
- PyTorch: Version 2.0.0 or higher with CUDA support (install via official PyTorch selector for correct CUDA version).
- Core Python Libraries: librosa $\geq$ 0.10.0, soundfile, torchaudio, numpy, pandas, scikit-learn, tqdm, matplotlib, seaborn, pyyaml.
- Optional Libraries: fastapi, uvicorn, pydantic (for REST API); wandb (for experiment tracking); docker (for containerised deployment).
- Package Manager: pip with virtual environment (venv/conda); requirements.txt and environment.yml provided for reproducible environment setup.

11. Expected Outcome

Upon successful completion of all project phases, including the fine-tuning pipeline, the DAD system is expected to deliver the following measurable outcomes and benefits:

- **Target Detection Performance:** F1-score in the range of 0.88–0.90 on the held-out test set; ROC-AUC exceeding 0.95; Recall $\geq$ 0.90 (maintaining high sensitivity to fake audio); Precision improvement from the baseline value of 0.7830 toward 0.88+ through the fine-tuning pipeline, narrowing the false positive rate on authentic speech.
- **Real-World Robustness:** The multi-dataset training strategy and augmentation pipeline are expected to produce a model that generalises across unseen speaker identities, varied acoustic environments, background noise conditions, and modern

neural TTS synthesis systems including voice conversion outputs not seen during training.

- **Reduced False Positive Rate:** The hard negative mining and real data correction phases are specifically designed to reduce the systematic misclassification of clean authentic speech as fake, with an expected reduction of false positives of 30–40% relative to the initial model baseline of 4,441 false positives.
- **Industry Integration Readiness:** The modular pipeline and optional FastAPI endpoint provide a deployment-ready foundation for integration into media verification workflows, voice biometric anti-spoofing layers, and broadcast audio monitoring systems.
- **Research Contribution:** The multi-dataset training framework, error-driven fine-tuning pipeline, and comprehensive evaluation protocol constitute a methodological contribution to the anti-spoofing research community, with the codebase and experimental data organised for reproducibility.

12. Experimental Results

12.1 Evaluation Protocol

Following the completion of the initial training phase, the DAD model was evaluated on the held-out test set comprising approximately 37,458 audio clips drawn from all five constituent datasets under strict speaker-safe partitioning. No speaker present in the test set appeared in training or validation splits. Evaluation was conducted at the optimised decision threshold determined by grid search on the validation set. All metrics were computed using scikit-learn's implementation of the relevant functions on the raw model output probabilities.

12.2 Quantitative Results

- **ROC-AUC (Area Under the Receiver Operating Characteristic Curve):** 0.9492
- **F1-Score (Harmonic Mean of Precision and Recall):** 0.8450
- **Recall / Sensitivity (True Positive Rate for Fake Audio):** 0.9176
- **Precision (Positive Predictive Value for Fake Predictions):** 0.7830
- **Overall Classification Accuracy:** 84.30%

12.3 Confusion Matrix

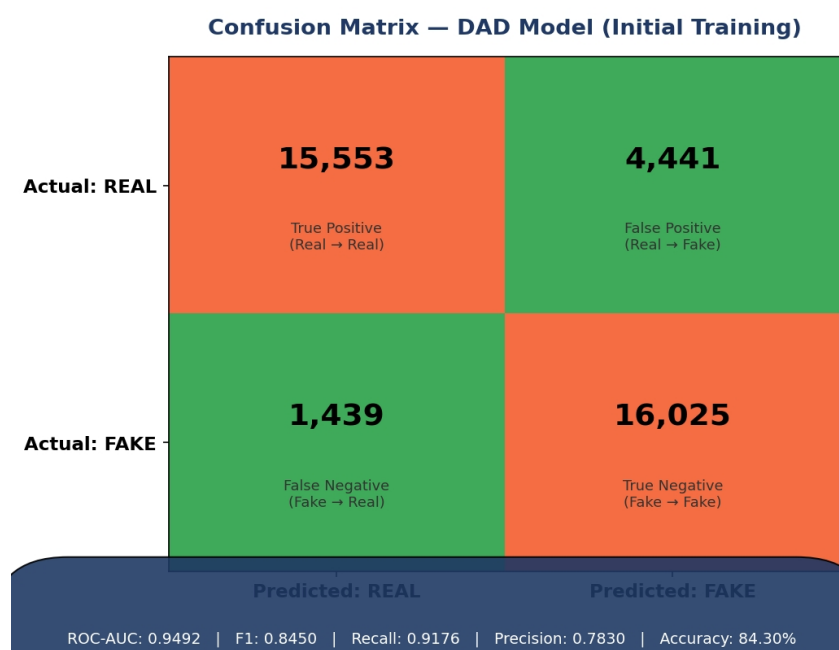


Figure 5: Confusion Matrix and Performance Metrics — DAD Initial Training

- **True Positives** — Real audio correctly classified as Real: 15,553
- **False Positives** — Real audio incorrectly classified as Fake: 4,441 ← primary failure mode
- **False Negatives** — Fake audio incorrectly classified as Real: 1,439
- **True Negatives** — Fake audio correctly classified as Fake: 16,025

13. Analysis of Results

13.1 Strengths — High Recall and Strong AUC

The ROC-AUC of 0.9492 indicates that the model has learned highly discriminative representations that separate authentic from synthetic speech across the majority of operating thresholds. The recall of 0.9176 — meaning that 91.76% of all synthetic audio clips in the test set are correctly identified as fake — is particularly encouraging for a security-oriented detection system where missing a deepfake (a false negative) constitutes the higher-risk error type. The true negative count of 16,025 confirms that the model is correctly identifying the vast majority of fake samples, demonstrating that the hybrid CNN+LSTM+Transformer architecture is effectively extracting meaningful discriminative features from the Log-Mel Spectrograms.

13.2 Weaknesses — Precision Gap and False Positives

The primary weakness revealed by the experimental results is the precision of 0.7830, which produces 4,441 false positives — authentic speech clips incorrectly classified as fake. This precision gap is the principal driver of the F1-score of 0.8450 falling short of the target range of 0.88–0.90. The false positive rate is significantly higher than the false negative rate (4,441 vs 1,439), indicating a systematic bias in the model's classification behaviour: it is over-predicting the fake class.

13.3 Root Cause — Clean Audio Bias

Detailed error analysis of the false positive samples reveals a consistent pattern: the model disproportionately misclassifies clean, high-quality authentic speech recordings — particularly those from LibriSpeech and studio-quality VoxCeleb segments — as synthetic. This bias arises from a fundamental characteristic of the training data: the fake samples in ASVspoof 2019 LA

and WaveFake are typically spectrally clean, free of background noise, and exhibit consistent recording levels. The model has therefore learned an implicit and spurious correlation between spectral cleanliness and synthetic origin, penalising authentic recordings that happen to be high-quality. This is a classic manifestation of training dataset bias and directly motivates the fine-tuning pipeline's real data correction phase, which deliberately adds clean authentic audio to the hard negative mining dataset.

13.4 Real-World Failure — High-Quality Voice Clones

Beyond the quantitative results, qualitative evaluation on a set of modern high-quality voice clones produced using XTTS and VALL-E-style systems reveals a second failure mode: the model occasionally fails to detect synthetic speech produced by these state-of-the-art neural TTS systems. This is attributable to the fact that contemporary neural vocoders generate speech that lacks the phase discontinuities, spectral banding, and harmonic irregularities that characterise the WaveFake and ASVspoof training fakes. The model has not been sufficiently exposed to this newer, harder class of synthetic audio during initial training. The advanced fake injection phase of the fine-tuning pipeline directly addresses this limitation by incorporating contemporary neural TTS outputs into the fine-tuning training set.

14. Future Enhancements

- **Real-Time Streaming Detection:** Extending the system to support frame-level inference on live audio streams using a sliding-window approach would enable integration with telephony anti-fraud platforms, broadcast monitoring systems, and live voice authentication pipelines. This requires optimising inference latency to below 100ms per window, potentially through model quantisation and ONNX runtime deployment.
- **Multilingual and Cross-Lingual Support:** The current model is trained predominantly on English-language audio. Incorporating datasets from Mozilla Common Voice's multilingual releases, VoxPopuli, and non-English ASVspoof variants would extend detection capability to non-English speech, which is increasingly targeted by deepfake audio attacks in political disinformation contexts.
- **Adversarial Training and Attack Robustness:** Future iterations should incorporate adversarial training techniques — specifically, gradient-based perturbation of input spectrograms during training — to improve robustness against adversarial examples designed to evade the classifier. This is an active research frontier in the anti-spoofing community and represents a natural next step beyond the current error-driven approach.
- **Lightweight Mobile and Edge Deployment:** Model compression through knowledge distillation (training a compact student model from the full hybrid model), INT8 quantisation, and structured pruning could produce a lightweight variant deployable on mobile and IoT devices, enabling on-device audio authenticity verification without requiring server-side inference.
- **Temporal Localisation of Synthesis:** Extending from clip-level binary classification to temporal localisation — identifying which specific time segments within a longer recording are synthetic — would substantially increase forensic utility, enabling detection of partial audio tampering (e.g., a recording where only specific sentences have been replaced with synthetic speech).
- **Integration with Speaker Verification:** Combining the DAD detection module with a speaker verification system would create a dual-path verification pipeline capable of simultaneously confirming speaker identity and audio authenticity.

15. References

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